Arista Access Node HEB-ACC-43 OSPF Adjacency Loss and BGP Route Churn Troubleshooting SOP

Node: HEB-ACC-43 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_2786

Description

The failure of OSPF adjacency establishment and BGP route churn on Arista Access Node HEB-ACC-43 at Hebbal Edge can be attributed to an underlying protocol issue or misconfiguration. In the context of OSPF, adjacency loss can occur due to various reasons such as mismatched OSPF process IDs, incompatible OSPF router types, or incorrect OSPF network statements. BGP route churn, on the other hand, might be a result of a BGP peering issue, BGP configuration discrepancies, or a network-wide routing policy conflict. The Arista Access Node is equipped with advanced routing capabilities, and its CLI provides a robust set of diagnostic tools to troubleshoot and resolve such issues. This SOP outlines the procedures to identify and rectify OSPF adjacency loss and BGP route churn on HEB-ACC-43, ensuring the network remains stable and optimized.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and that the access node is powered on.

- Check the status of the redundant links and ensure that they are operational.

- Dispatch remote hands to Hebbal Edge to perform physical checks and verify the node's hardware status.

Potential Root Causes

Cause	Probability
Mismatched OSPF process IDs	High
Incompatible OSPF router types	Medium
Incorrect OSPF network statements	Low
BGP peering issue	High
BGP configuration discrepancies	Medium
Network-wide routing policy conflict	Low

Diagnostic Commands & Expected Outputs

```
Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command displays the
OSPF neighbor states, helping identify adjacency loss issues.

Command: show bgp summary, Expected Output: BGP router identifier
192.0.2.1, local AS number 65000, Description: This command provides
an overview of the BGP session status, aiding in the diagnosis of BGP
```

```
route churn.

Command: show ip bgp neighbor, Expected Output: Neighbor 192.0.2.2
Remote AS 65001 Update-group 1, Description: This command shows
detailed information about the BGP neighbor status, facilitating the
identification of peering issues.
```

Resolution Steps

Step 1: Step: Verify OSPF process IDs and router types, Command: show ospf database, Description: Use this command to verify that the OSPF process IDs and router types are correctly configured.

Step 2: Step: Update OSPF network statements, Command: config t router ospf 100 network 10.0.0.0/24, Description: Update the OSPF network statements to resolve any discrepancies, ensuring correct OSPF adjacency establishment.

Step 3: Step: Re-establish BGP peering, Command: clear bgp peer-group, Description: Use this command to re-establish BGP peering with the affected neighbor, resolving any peering issues.

Step 4: Step: Verify BGP configuration, Command: show bgp config, Description: Verify that the BGP configuration is correct and free from discrepancies.

Post-Check Verification

```
Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: Verify that the OSPF
adjacency is established.

Command: show bgp summary, Expected Output: BGP router identifier
192.0.2.1, local AS number 65000, Description: Verify that the BGP
session is stable and free from route churn.
```

Roll-back Procedures

- Step: Restore previous OSPF configuration, Command: show run | inc ospf, Description: Restore the previous OSPF configuration to its original state.

- Step: Re-establish BGP peering, Command: clear bgp peer-group, Description: Re-establish the BGP peering with the affected neighbor.